

The Physics of Prospective Learning

Prospective State, Phase-Gradient Carriers, and a Failure-Capable Test Program

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Abstract

An intelligent system can recognize the present correctly and still fail to act intelligently if it cannot maintain what is changing, where that change may lead, which actions remain available, and when new evidence should revise its plan. This is the prospective-state problem. Existing work provides important parts of the answer through predictive processing, active inference, recurrent planning, traveling waves, oscillatory routing, fast weights, and test-time adaptation. What remains open is how these parts can be joined into a measurable state-maintenance architecture that operates while the future is still unfolding.

This paper defines a prospective state as a coupled set of current context, trajectory estimate, candidate future policies, working state, and receiver-relative mismatch. It then proposes phase gradients as one candidate physical carrier of trajectory information. A local phase field supplies measurable spatial and temporal derivatives; the receiver-relative difference between an expected phase relation and an arriving one supplies a candidate update signal. In Self Aware Networks (SAN), that more specific operation is called a **Phase Wave Differential (PWD)**. The term is introduced only after the underlying problem and operation are defined: it is not a synonym for phase coding or traveling waves.

The paper formalizes phase velocity, circular mismatch, gated state revision, fast and slow state variables, and the conditions under which coherent population summation can improve signal-to-noise ratio. It retains the original prospective-learning program - dynamic routing, counterfactual phase configurations, reset amplification, micro-to-macro amplification, a slow context carrier, a fast inference stream, and phase-to-structure consolidation - while separating established evidence, engineering analogies, hypotheses, and open obligations. Six experiments test whether phase-derived features add out-of-sample predictive value beyond firing-rate, recurrent-state, motion-energy, attention, replay, and predictive-coding baselines.

1. The Ordinary Problem: Acting Before the Next Complete Observation

Imagine reaching for a moving cup while another person is still sliding it across a table. The image at the retina is already delayed. The hand cannot wait for a complete future observation because, by then, the cup will have moved again. The nervous system must maintain a provisional trajectory, prepare one or more actions, and revise those preparations as new evidence arrives.

The same problem appears in an artificial agent. A system may correctly describe its current screen yet repeatedly lose its goal, reconstruct its plan from scratch, or continue an obsolete plan after the environment changes. Present-state recognition is not enough. The system needs a state that is:

- persistent enough to preserve direction and purpose;
- revisable enough to absorb surprise;
- structured enough to compare candidate futures;
- measurable enough to distinguish the proposal from ordinary recurrence or attention.

Kording, Vogelstein, Chaudhari, and Verstynen describe prospective learning as a needed science of learning under future change rather than only from a fixed past distribution [1]. This paper asks a

narrower mechanistic question: **what state must persist while the future remains incomplete, and what measurable dynamics could carry and update that state?**

2. What Established Accounts Already Explain

Prospective cognition does not begin from an empty scientific landscape.

Predictive-coding models explain how top-down predictions and bottom-up residuals can be exchanged in hierarchical circuits [2]. The free-energy and active-inference framework relates perception, action, uncertainty, and expected trajectories within a generative-model account [3]. Recurrent-network models can generate planning sequences and replay that predict aspects of human behavior [4]. Traveling-wave studies show that phase structure propagates across cortex and is shaped by anatomy [5-7]. Communication-through-coherence proposes that phase relations can regulate effective communication between neural populations [8], while frequency-resolved work associates alpha / beta and gamma activity with different directions of cortical influence [9].

Artificial systems provide complementary engineering precedents. Fast weights retain recent structure without rewriting all durable parameters [10]. Test-time training changes a system during inference under distribution shift [11]. Prioritized replay selects high-value errors or experiences for later learning [12].

These accounts establish plausible components. They do not, by themselves, establish this paper’s joined proposal. In particular:

1. a traveling wave describes propagation but does not automatically define a future-oriented state;
2. phase coding associates information with phase but does not automatically specify a receiver-relative update operation;
3. recurrence preserves activity but does not automatically distinguish useful persistence from perseveration;
4. prediction error identifies mismatch but does not automatically identify which physical variable carries that mismatch;
5. test-time adaptation demonstrates engineering utility but does not identify a biological mechanism.

The contribution here is a model-and-test architecture joining those open interfaces.

3. Define the Prospective State Before Naming Its Carrier

Let a prospective state at time t contain:

$$\mathcal{S}_t = (C_t, T_t, \mathcal{H}_t, P_t, W_t, F_t, G_t). \quad (1)$$

The terms have operational roles:

Term	Plain-language role	Candidate observable or implementation
C_t	Current context	sensory state, task state, recurrent context
T_t	Trajectory estimate	motion estimate, policy rollout, hidden-state derivative
$\mathcal{H}_t$	Candidate futures or policies	planning branches, replayed paths, active hypotheses
P_t	Persistent plan or mission state	goal stack, sustained population state, durable agent memory
W_t	Fast working state	transient inference buffer, short-window neural state

Term	Plain-language role	Candidate observable or implementation
F_t	Mismatch or frustration	prediction residual, conflict score, phase mismatch
G_t	Revision gate	update probability, reset signal, inhibitory or compute-control event

Equation (1) is a state definition, not a claim that one brain region stores a literal tuple. It gives an experimenter or system designer a list of functions that must be measured or implemented.

The transition problem is:

$$\mathcal{S}_{t+1} = \mathcal{U}(\mathcal{S}_t, y_{t+1}, a_t), \quad (2)$$

where y_{t+1} is new evidence, a_t is the action taken, and $\mathcal{U}$ is an update operation. A successful prospective learner preserves useful trajectory information across this transition without freezing the state against contradictory evidence.

4. The Physical Candidate: A Phase Field With Direction and Rate

Suppose a measured neural population supports a local phase field $\phi(\mathbf{x}, t)$. For a narrow-band component, its local spatial and temporal derivatives are:

$$\begin{aligned} \mathbf{k}(\mathbf{x}, t) &= \nabla \phi(\mathbf{x}, t), \\ \omega(\mathbf{x}, t) &= -\partial_t \phi(\mathbf{x}, t). \end{aligned} \quad (3)$$

For a plane wave $\phi = \mathbf{k} \cdot \mathbf{x} - \omega t + \phi_0$, constant-phase surfaces satisfy:

$$\partial_t \phi + \mathbf{v}_\phi \cdot \nabla \phi = 0. \quad (4)$$

The minimum-norm velocity normal to those phase surfaces is therefore:

$$\mathbf{v}_\phi = -\frac{\partial_t \phi}{\|\nabla \phi\|^2} \nabla \phi = \frac{\omega}{\|\mathbf{k}\|^2} \mathbf{k}. \quad (5)$$

In one spatial dimension, Equation (5) reduces in magnitude to $v_\phi = \omega/k$. This correction matters: the familiar scalar relation is not a general vector law, and it becomes ill-conditioned where the measured spatial gradient approaches zero.

A phase gradient can therefore carry a measurable direction and rate. That does not prove that the brain uses the field prospectively. The scientific proposal is narrower: phase-derived features may contribute incremental information about T_t , routing, or update timing before the next complete sensory state is available.

5. The Receiver-Relative Operation and Why It Needs a New Name

An arriving oscillatory event does not meet an empty receiver. The receiving population already has a phase, membrane state, recent history, inhibition, and an expected relation to upstream activity. Define the wrapped receiver-relative difference:

$$\Delta\phi_j(t) = \text{wrap}_{(-\pi, \pi]} \left[\phi_j^{\text{in}}(t - \tau_j) - \hat{\phi}_j^{\text{recv}}(t) \right], \quad (6)$$

where τ_j is a measured or estimated route delay. A weighted circular mismatch is:

$$F_\phi(t) = \frac{\sum_j w_j [1 - \cos \Delta\phi_j(t)]}{\sum_j w_j}, \quad (7)$$

$$0 \leq F_\phi \leq 2.$$

Equation (7) converts a collection of phase relations into a bounded candidate frustration signal. Small values mean that the arriving and expected relations are phase-compatible under the chosen delays; large values mean that they differ.

Only now is the SAN term useful. A **Phase Wave Differential** is the candidate receiver-relative difference operation represented by Equation (6), together with its measurable consequence for the receiving system. The distinction from inherited vocabulary is functional:

- **phase coding** says that information can be associated with phase;
- **traveling-wave analysis** estimates how phase propagates across space;
- **predictive coding** compares predictions with sensory evidence at an algorithmic level;
- **PWD** proposes a receiver-relative physical difference between an arriving phase relation and the receiver's current or expected relation, with that difference available to alter routing, recruitment, learning, or action.

The concept will be scientifically distinct only if PWD-derived variables explain or predict something beyond the older descriptions. If Equation (6) adds no incremental value after phase, amplitude, firing rate, recurrence, attention, and prediction error are controlled, the new term has failed its discrimination test.

6. Mismatch Must Control Revision Without Erasing Continuity

A revision gate can be written schematically as:

$$G_t = \sigma(\alpha F_t + \beta U_t + \gamma Q_t - \theta), \quad (8)$$

where U_t represents uncertainty, Q_t represents task conflict or resource pressure, θ is a threshold, and σ is a bounded activation function. This is not a claim that a biological gate uses a logistic function. It states the control requirement: mismatch and uncertainty should change the probability or magnitude of state revision.

The fast working state can respond quickly:

$$\tau_W \dot{W} = -W + f_W(C_t, T_t, \mathcal{H}_t, F_t, G_t), \quad (9)$$

while the persistent plan state changes more slowly:

$$\begin{aligned}\tau_P \dot{P} &= -P + f_P(P_t, W_t, a_t, y_{t+1}), \\ \tau_P &\gg \tau_W.\end{aligned}\tag{10}$$

A discrete gated update makes the stability-plasticity tradeoff explicit:

$$P_{t+1} = (1 - G_t)P_t + G_t \tilde{P}_{t+1},\tag{11}$$

where $\tilde{P}_{t+1}$ is the candidate revised plan. When G_t is too small, the agent perseverates. When it is too large, the agent repeatedly discards useful continuity. Prospective learning requires a measurable operating region between those failures.

7. Dynamic Routing and Counterfactual Futures

The original prospective-learning manuscript proposed that synaptic structure stores a map of possible routes while changing phase relations help select which route is functionally available. Communication-through-coherence and frequency-dependent feedforward / feedback findings make phase-sensitive routing biologically plausible [8,9], but they do not establish this particular prospective operator.

The retained hypothesis is:

1. P_t constrains which candidate futures belong to $\mathcal{H}_t$;
2. W_t instantiates one or more candidates transiently;
3. phase relations affect which populations exchange effective influence;
4. receiver-relative mismatch contributes to F_t ;
5. G_t determines whether the plan is maintained, revised, or reset;
6. action and returned sensory evidence test the selected trajectory.

Counterfactual simulation is the strongest and riskiest extension. The proposal is not that every imagined future is literally a separate wave. It is that candidate actions may be accompanied by distinguishable transient phase configurations, and that their relation to learned recurrent structure may predict which action is selected. Recurrent planning and replay are established alternatives [4]. The phase account earns additional explanatory status only if its features improve held-out prediction beyond replay content and firing-rate state.

8. Fast Phase Adaptation and Slow Structural Consolidation

The historical paper described a multi-frequency architecture using the metaphor of a slow **canvas** and fast **ink**. In current operational language:

- a slower or more persistent state supplies context and constraints;
- faster activity carries transient evidence and local inference;
- repeated successful state transitions can influence slower structural learning.

This does not require one fixed frequency band to own context across all tasks. Delta, theta, alpha, beta, gamma, aperiodic activity, and non-oscillatory recurrent states can change roles with anatomy and task. The measurable claim is about timescale separation, not a universal frequency dictionary.

The two-stage memory hypothesis retains the original intent:

Timescale	Candidate operation	Failure if absent
Fast	reversible phase or state reconfiguration	slow response to changing context
Slow	consolidation of repeatedly successful trajectories	useful changes vanish after the episode

Fast weights and test-time adaptation provide engineering analogies [10,11]; prioritized replay provides a comparator for selecting informative errors [12]. In biology, any claim that phase patterns bias long-term plasticity must be tested against co-activity, firing rate, neuromodulation, sleep / replay, and established synaptic-plasticity variables. Phase persistence alone is not evidence of consolidation.

9. Reset Amplification and Its Physical Conditions

The original paper used inhibitory reset to explain how a small timing difference might influence a larger population. Optogenetic work shows that driving fast-spiking interneurons can induce gamma rhythmicity and alter sensory responses [13]. That supports an inhibitory-rhythm interaction, not the full prospective-reset hypothesis.

If N coupled units each contribute an aligned signal amplitude a , while independent zero-mean noise has standard deviation σ , then:

$$\text{SNR}_{\text{ind}} = \frac{Na}{\sigma\sqrt{N}} = \frac{a}{\sigma}\sqrt{N}. \quad (12)$$

This square-root gain requires aligned signal and approximately independent noise. With mean pairwise noise correlation ρ , the summed noise standard deviation becomes:

$$\begin{aligned} \sigma_N &= \sigma\sqrt{N[1 + (N-1)\rho]}, \\ \text{SNR}_\rho &= \frac{Na}{\sigma_N}. \end{aligned} \quad (13)$$

Positive correlated noise can sharply reduce or saturate the gain. Reset amplification is therefore conditional on coupling, selectivity, alignment, noise covariance, and timing. A broad reset that erases useful trajectory information is a failure, not amplification.

The micro-to-macro point is retained separately. Rod photoreceptors can produce a measurable electrical response following a single-photoisomerization event [14]. This demonstrates biological amplification of microscopic triggers. It does not validate PWD, long-lived quantum coherence, or prospective phase dynamics. The analogy establishes possibility of scale transfer, not identity of mechanism.

10. A Bounded Engineering Mirror for Artificial Agents

The biological and artificial cases share a control problem without requiring substrate identity.

An agent needs:

- a persistent mission or plan state P_t ;
- a fast scratch or inference state W_t ;
- a mismatch signal F_t ;

- a gate G_t deciding when to revise or spend resources;
- a trajectory estimate T_t that survives across local operations.

Preserving every prior thought can lock the agent into obsolete reasoning. Reconstructing the plan from scratch can produce drift, repetition, and goal loss. Equation (11) names the middle problem.

The original manuscript discussed “preserved thinking” in a specific model workflow. That material remains part of the private development genealogy, not public evidence about a commercial system. The public claim is architecture-level: persistent and fast states plus controlled revision are testable design requirements. MASI supplies a separate engineering analogy in which independently measured optical wavefields are computationally phase-synchronized and fused [15]. It demonstrates that distributed phase alignment can solve a reconstruction problem in engineered optics. It does not show that neural tissue uses MASI, NAPOT, or the prospective mechanism proposed here.

11. Six Decisive Tests

Every central claim must compete with simpler explanations.

Test 1: Does phase structure improve trajectory decoding?

Measure spatial phase, temporal phase, amplitude, firing rate, recurrent latent state, and behavior during motion extrapolation. Compare held-out models:

$$\Delta\mathcal{L}_\phi = \mathcal{L}_{\text{base}+\phi}^{\text{test}} - \mathcal{L}_{\text{base}}^{\text{test}}, \quad (14)$$

where $\mathcal{L}$ is a preregistered predictive score. The phase-gradient hypothesis requires a reproducible positive increment after matched complexity and leakage controls. If firing rate, motion energy, and recurrent state explain the same variance, the proposed carrier is not supported.

Test 2: Is task-switch mismatch receiver-relative?

At task switches, estimate route delays and compute Equation (6). Compare PWD-derived mismatch with ordinary phase, amplitude, event-related potentials, reaction time, and attention-gating models. The claim fails if receiver-relative mismatch is redundant or unstable across sessions.

Test 3: Do candidate futures have predictive phase configurations?

During planning, decode candidate actions from replay content, firing rate, and phase configuration. The counterfactual extension requires phase-derived information to predict eventual choice beyond replay and sequence-model baselines. Mere pre-action oscillation is insufficient.

Test 4: Does a reset improve revision rather than merely mark surprise?

Measure inhibitory recruitment, population phase dispersion, surprise, and subsequent adaptation. A prospective reset should predict faster or more accurate model revision after controlling for evoked responses, arousal, movement, and attentional reorientation. If reset markers track surprise but not revision, the proposed causal role is unsupported.

Test 5: Does phase history add to plasticity prediction?

Measure repeated phase relations, spike timing, firing rate, neuromodulation, and later structural or behavioral retention. The phase-to-structure hypothesis requires an incremental contribution beyond conventional plasticity variables. Correlation between coherence and later learning is not enough.

Test 6: Does the two-timescale architecture outperform one-state controls?

In artificial agents, compare a two-state gated architecture with:

- a recurrent single-state model;
- a larger context window;
- retrieval-augmented memory;
- fast weights;
- test-time training;
- plan reconstruction at every step.

Use matched parameter count, compute, and information access. Measure goal retention, adaptation latency, catastrophic drift, perseveration, calibration, and energy or compute cost. The architecture fails if a simpler control matches its performance.

12. Claim Classes, Boundaries, and Failure Conditions

Statement	Current class	What would strengthen it	What would defeat it
Prospective behavior requires maintained future-oriented state	operational architecture	cross-task measurement and intervention	equally capable stateless control
Phase gradients can carry direction and rate	mathematical / measurement fact	robust neural estimation	gradients too unstable or ill-defined
Phase gradients carry prospective trajectory information	mechanism hypothesis	incremental held-out decoding	no value beyond baseline
PWD is a receiver-relative update variable	SAN mechanism proposal	intervention on predicted route relation	redundancy with simpler variables
Phase configurations support counterfactual futures	speculative / testable extension	phase-specific choice prediction and perturbation	replay / rate models fully explain result
Inhibitory reset amplifies useful phase structure	conditional mechanism	selective reset improves later revision	reset only suppresses or erases
Fast phase state biases slow consolidation	hypothesis	incremental plasticity prediction	no effect beyond co-activity / rate
MASI supports feasibility of distributed phase fusion	engineering analogy	none needed for the analogy	cannot validate neural identity
Single-photon transduction shows micro-to-macro amplification	biological precedent	none needed for precedent	cannot validate PWD or prospection

The paper does not claim that phase dynamics replace predictive coding, active inference, replay, recurrent computation, or synaptic learning. It claims that receiver-relative phase dynamics may supply a measurable implementation layer that can be tested against those accounts.

13. Lossless Version Genealogy

The version history matters because later caution should refine, not erase, the original research program.

1. **Historical Version 4 (December 2025).** The original paper joined phase gradients, PWD, dynamic routing, counterfactual simulation, reset amplification, single-photon amplification, slow context / fast inference, test-time adaptation, redness as an illustrative attractor, and phase-to-structure memory [16].
2. **Program Edition (July 2026).** The first program edition narrowed the claims, defined P, W, G, F, T , added baseline-dependent predictions, and separated companion-paper ownership [17].
3. **This final-preprint successor.** The present version preserves both layers. It corrects the vector phase-velocity relation, adds a circular mismatch model, a gated two-timescale architecture, correlated-noise limits, complete citation use, earned vocabulary, and explicit disposition of every major historical proposition.

The illustrative “redness” example remains part of the genealogy, but this paper does not use it as evidence. A stable percept may be modeled as an attractor or recurrent state in other SAN work; proving that it is a PWD pattern requires direct measurement and belongs to the dedicated rendering and qualia program.

Neural Array Projection Oscillation Tomography (NAPOT) also remains a historical companion hypothesis. This paper requires only the possibility that phase-derived variables contribute to trajectory maintenance. It does not require, and therefore does not establish, the full NAPOT rendering operator.

14. Open Scientific Obligations

The program remains incomplete until it supplies:

1. a preregistered operational definition of phase, delay, receiver, and expected relation;
2. a dataset with spatial sampling adequate to estimate Equation (5);
3. leakage-safe out-of-sample comparisons against all named baselines;
4. interventions that distinguish phase consequence from common drive;
5. a model-selection penalty for the added phase variables;
6. replication across subjects, tasks, and recording modalities;
7. a clear result for when phase information fails;
8. an explicit bridge, if any, from fast phase state to measured structural plasticity.

No empirical result is reported in this paper. The equations are a measurement and falsification scaffold, not a fitted theory of the brain.

Conclusion

Prospective intelligence requires more than recognizing the present. It requires a maintained yet revisable state that carries context, trajectory, candidate futures, mismatch, and control over revision. This paper defines that state before naming a mechanism.

Phase gradients are a physically measurable candidate carrier because their spatial and temporal derivatives can encode direction and rate. A Phase Wave Differential then names the more specific SAN proposal: the receiver-relative difference between an arriving phase relation and the relation already present or expected at the receiver, when that difference has a measurable downstream consequence. The proposal is not secured by resemblance to phase coding, traveling waves, predictive coding, or engineered phase synchronization. It succeeds only if its variables add predictive and causal power beyond those alternatives.

The result is a stronger and more faithful research program: preserve the original architecture, expose its assumptions, correct its mathematics, and make every major proposition capable of losing.

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